Last Time:

- Quaternion Algebra
- Quaternion Kinematics
- Quaternion Exponential

Today:

- Mid-Term Review
- Newton-Euler Dynamics
- Quadrotor

Mid-Term Logistics:

- Thursday 10/29 in class (33-225)
- Bring one page of notes (Front + Back)
- Nothing else (no calculators or computers etc.)
- If you have a documented accomodation for more time and/or reduced distraction, we'll have an additional room (31-201M) from 10:00 - 2:00. You can start early or stay late.
- TAs in both rooms
- Make-Up Time: Monday (Nov. 2) 6:30 - 8:00 PM

Mid-Term Topics:

- PSets 1-4 + Quizes 1-4
- Lectures 1-9

* Newton's 2nd law in Cartesian + Polar

* Kinetic + Potential Energy in arbitrary coordinates

$$T = \underbrace{\frac{1}{2} m {}^N v^T {}^N v}_{\text{particle}} + \underbrace{\frac{1}{2} {}^B\omega^T J {}^B\omega}_{\text{rigid-body rotation}}$$

- Make sure to map $\dot{q} \to {}^N v$ and ${}^B\omega$

$$U = \underbrace{mg}_{\text{gravity}} \underbrace{h(q)}_{\text{height}} + \underbrace{\frac{1}{2} k}_{\text{spring}} \underbrace{d(q)^2}_{\text{displacement}}$$

* Dynamics from Euler-Lagrange Equation!

$$L = T - U, \quad \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = \underbrace{F_i}_{\text{generalized force}}$$

* Manipulator Equation + mass matrix:

$$\underbrace{M(q)}_{\text{Mass Matrix}} \ddot{q} + \underbrace{C(q,\dot{q})}_{\text{coriolis}} + \underbrace{G(q)}_{\text{Gravity/Potential}} = F \leftarrow \text{generalized force}$$

$$M(q) > 0 \Rightarrow v^T M(q) v \geq 0 \ \forall v \Rightarrow \text{eigvals}(M) > 0$$

* Momentum

$$p = \frac{\partial L}{\partial \dot{q}} = \frac{\partial T}{\partial \dot{q}} = M(q)\dot{q}$$

* Rotating reference frames + velocity transport theorem

$${}^N r = Q\,{}^B r \Rightarrow {}^N \dot{r} = Q\,{}^B \dot{r} + \dot{Q}\,{}^B r = Q\,{}^B \dot{r} + Q S({}^B\omega)\,{}^B r$$
$$= Q\left[{}^B \dot{r} + {}^B\omega \times {}^B r\right]$$

* Work done by a force + power:

$$W = \Delta T = \int_{t_0}^{t_f} \dot{T}\, dt = \int_{t_0}^{t_f} \underbrace{\vec{F} \cdot \vec{v}}_{\text{Power}}\, dt = \int_{\vec{r}_0}^{\vec{r}_f} \vec{F} \cdot d\vec{r}$$

$$T = \frac{1}{2} m v^T v \Rightarrow \dot{T} = \frac{\partial T}{\partial v}\dot{v} = m v^T \dot{v} = v^T (m\dot{v}) = v^T F$$

- Constraint forces do no work $\Rightarrow \vec{F}_c \cdot \vec{v} = 0$

* Rigid-Body Inertia:

$$J > 0 \Rightarrow \omega^T J \omega \geq 0 \quad \forall \omega, \; \text{eigvals}(J) > 0$$

* Parallel-Axis Theorem:

- Calculate CoM + inertia of composite bodies

* Rigid-Body Dynamics:

- Euler's equation: $J\dot{\omega} + \omega \times J\omega = \tau$

- Equilibria + stability
- Energy + momentum
- Rotation matrices but no quaternions

* Linearization

- Nonlinear ODE → linear approximation
- 2nd-order / Quadratic → linear ODE

- Natural frequencies

* Stability:

- Local / Linear stability from eigenvalues and/or exponential solutions

$\dot{x} = ax$ (a: scalar) $\Rightarrow x(t) = e^{at} x_0$

$a > 0 \Rightarrow$ unstable (blows up)
$a < 0 \Rightarrow$ stable (decays)

$\dot{x} = Ax \Rightarrow x(t) = e^{At} x_0$

(A: matrix; e^{At}: matrix exponential)

$$e^{At} = \sum_k \frac{1}{k!} A^k \approx I + A + \frac{1}{2} A^2 + \frac{1}{6} A^3 \cdots$$

$A = V D V^T$, $V^T V = I$

$D = \begin{bmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \ddots \lambda_n \end{bmatrix}$

$\Rightarrow e^{At} = e^{VDV^T t} \approx I + VDV^T + \frac{1}{2} \underbrace{VDV^T VDV^T}_{VD^2V^T} + \cdots$

$\Rightarrow V e^{Dt} V^T$

$\Rightarrow$ Decouples into n scalar systems in eigen basis:

$y = V^T x \Rightarrow y_i(t) = e^{\lambda_i t} y_i(0)$

$\Rightarrow$ Check stability by looking at real part of each λ_i

if $Re[eigvals(A)] < 0 \Rightarrow$ stable

if any $Re[eigvals(A)] > 0 \Rightarrow$ unstable

if any " $= 0 \Rightarrow$ inconclusive

* Lyapunov Stability from energy:

$$x = \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix}, \quad V(x) > 0 \quad \forall x \neq 0$$

$$\dot{V}(x) = \frac{\partial V}{\partial x}\dot{x} = \frac{\partial V}{\partial x} f(x) \leq 0 \quad \text{"Lyapunov stable"}$$

$$\dot{V}(x) < 0 \quad \text{"asymptotically stable"}$$

Exam Format:

- 2 long-form problems
  - * Lagrangian Dynamics 35%
  - * Rigid body dynamics 35%
- ~10 short-answer and/or multiple choice. 30%

Newton-Euler Dynamics:

- Combine translation + Attitude dynamics for a rigid-body system
- Super common: e.g. airplanes, drones, rocket, satellites, quadrupeds, etc.

- Common choice for state:

$$x = \begin{bmatrix} {}^N r \\ {}^N q^B \\ {}^N v \\ {}^B \omega \end{bmatrix} \Rightarrow \dot{x} = \begin{bmatrix} {}^N \dot{r} \\ \dot{q} \\ {}^N \dot{v} \\ {}^B \dot{\omega} \end{bmatrix} = \underbrace{\begin{bmatrix} {}^N v \\ \frac{1}{2} G(q)\, \omega \\ \frac{1}{m} {}^N F \\ J^{-1}\left({}^B \tau - {}^B \omega \times J\, {}^B \omega\right) \end{bmatrix}}_{f(x)}$$

- Another common choice:

$$x = \begin{bmatrix} {}^N r \\ {}^N q^B \\ {}^B v \\ {}^B \omega \end{bmatrix} \leftarrow \text{velocity in B frame}$$

- Translation Kinematics:

$${}^N \dot{r} = {}^N v = Q\, {}^B v$$

- Newton's 2nd law in Body frame:

$$\frac{1}{m} {}^N F = {}^N \dot{v} = \dot{Q}\, {}^B v + Q\, {}^B \dot{v} = Q\, S(\omega)\, {}^B v + Q\, {}^B \dot{v}$$

$$\Rightarrow {}^B \dot{v} = Q^T\, {}^N \dot{v} - {}^B \omega \times {}^B v$$

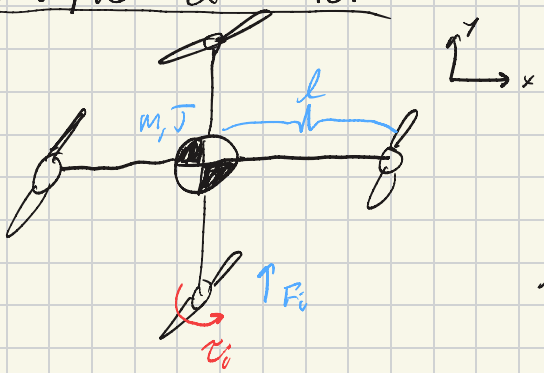
$$= Q^T \left(\frac{1}{m} {}^N F\right) - {}^B \omega \times {}^B v$$

$$= \boxed{\frac{1}{m} {}^B F - {}^B \omega \times {}^B v}$$

$$\Rightarrow \dot{x} = \begin{bmatrix} Q\, {}^B v \\ \frac{1}{2} G(q)\, {}^B \omega \\ \frac{1}{m} {}^B F - {}^B \omega \times {}^B v \\ J^{-1}\left({}^B \tau - {}^B \omega \times J\, {}^B \omega\right) \end{bmatrix}$$

* Simulating different systems is now just a matter of plugging in the right F and τ

Example: Quadrotor



$$F_i = K_F u_i$$

u_i: motor command

K_F: Thrust coefficient

$$\tau_i = K_\tau u_i$$

K_τ: Torque coefficient

$$^{B}F = Q^T \begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ K_F & K_F & K_F & K_F \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix}$$

$$^{\theta}\tau = \begin{bmatrix} \ell K_F (u_2 - u_4) \\ \ell K_F (u_3 - u_1) \\ K_\tau (u_1 - u_2 + u_3 - u_4) \end{bmatrix}$$

"$\tau = r \times F$"